THE TENSOR FUNCTOR WITH A DG BIMODULE IS A FUNCTOR OF DG CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-tensor (source lines 4944-4963); the proof environment begins two lines below it and is its own proof, closing at source line 4995.

Editorial note. The statement invokes the author's lemma lemma-functorial by label; that lemma's statement and its own proof are retained as a context span placed before the counted statement, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The proof is a signed computation on decomposable tensors; the align environment displaying the cancellation is part of the retained source range and is reproduced as written.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: tensoring with a dg bimodule is a functor of dg categories and induces functors on the module and homotopy categories.

Allowed standard prerequisites (editorial).

- Tensor product of dg modules over a dg algebra. For a dg algebra A and dg modules M over A and N over A, the tensor product M tensor_A N is a dg module whose differential satisfies the signed Leibniz rule on decomposable tensors.. Both displayed differentials and their expansions rest on that rule (source lines 4974-4989).
- Differential of a homogeneous module homomorphism. For homogeneous f : M -> M' of degree n the differential in the Hom complex is d of M' composed with f minus the signed composition of f with d of M.. It is the shape of the differential of f and of f tensor the identity, which the proof compares (source lines 4970-4981).
- Induced functors on the module and homotopy categories. A functor of dg categories compatible with differentials induces functors between the module categories and their homotopy categories (the author's lemma lemma-functorial, retained as context).. It is invoked in the statement to pass from the dg functor to the functors on Mod and K(Mod) (source line 4962).

Lemma 0.1. *Let R be a ring. A functor $F : \mathcal{A} \to \mathcal{B}$ of differential graded categories over R induces functors $Comp(\mathcal{A}) \to Comp(\mathcal{B})$ and $K(\mathcal{A}) \to K(\mathcal{B})$.*

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Proof. Omitted. □

Lemma 0.2. *Let R be a ring. Let (A, d) and (B, d) be differential graded algebras over R. Let N be a differential graded (A, B)-bimodule. Then $M \mapsto M \otimes_A N$ defines a functor*

$$- \otimes_A N : Mod^{dg}_{(A,d)} \longrightarrow Mod^{dg}_{(B,d)}$$

of differential graded categories. This functor induces functors

$$Mod_{(A,d)} \to Mod_{(B,d)} \quad and \quad K(Mod_{(A,d)}) \to K(Mod_{(B,d)})$$

by an application of Lemma 0.1.

Proof. Above we have seen how the construction defines a functor of underlying graded categories. Thus it suffices to show that the construction is compatible with differentials. Let M and M' be differential graded A-modules and let $f : M \to M'$ be an A-module homomorphism which is homogeneous of degree n. Then we have

$$\mathrm{d}(f) = \mathrm{d}_{M'} \circ f - (-1)^n f \circ \mathrm{d}_M$$

On the other hand, we have

$$\mathrm{d}(f \otimes \mathrm{id}_N) = \mathrm{d}_{M' \otimes_A N} \circ (f \otimes \mathrm{id}_N) - (-1)^n (f \otimes \mathrm{id}_N) \circ \mathrm{d}_{M \otimes_A N}$$

Applying this to an element $x \otimes y$ with $x \in M$ and $y \in N$ homogeneous we get

$$\begin{aligned} \mathrm{d}(f \otimes \mathrm{id}_N)(x \otimes y) =& \mathrm{d}_{M'}(f(x)) \otimes y + (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ & - (-1)^n f(\mathrm{d}_M(x)) \otimes y - (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ =& \mathrm{d}(f)(x \otimes y) \end{aligned}$$

Thus we see that $\mathrm{d}(f) \otimes \mathrm{id}_N = \mathrm{d}(f \otimes \mathrm{id}_N)$ and the proof is complete. □